MIL-C-46876(MI)
14 December 1973
SUPERSEDING
MIS 13019
18 February 1965

MILITARY SPECIFICATION

CEMENT, CARBONACEOUS

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 Scope. This specification covers one type of carbonaceous cement used in rocket motors.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

RR-S-366	Sieve, Test

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with a specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

FSC-5640

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

PUBLICATIONS

American Society for Testing and Materials

ASTM D 2240 Indentation Hardness of Rubber and Plastics by Means of a Durometer, Test for,

(Application for copies should be addressed to the American Soceity for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The material shall be a two part cement system composed of carbon solid and furfural (aldehyde liquid).

3.2 Chemical and physical properties. The chemical and physical properties shall be as specified in Table I.

Table I

Chemical and Physical Properties

Property	Minimum	Maximum	Test Method
Hardness, Shore D	75	90	4.3.2
Non-Extractables (of powder), percent	79.0	83.0	4.3.4
Screen analysis (of powder), percent passing through a U.S. Number 200 sieve	60	--	4.3.5

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3.3 Workmanship. The material supplied under this specification shall be uniform in quality and shall be free from impurities and other defects that would prevent its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. All inspections specified herein are classified as quality conformance inspections.

4.2.1 Inspection lot. An inspection lot shall consist of material of a single batch subjected to the same processing operations and under essentially the same conditions and presented for inspection at one time.

4.2.2 Sampling. Sampling for visual examination shall be 100 percent of the lot. Failure of any sample to meet the requirements of 3.3 and Section 5 shall cause rejection of the sample. Sampling for other examinations and tests shall be in accordance with MIL-STD-105, Level II, single sampling, acceptable quality level (AQL) 4.0 percent.

4.2.3 Procedure. Samples taken in accordance with 4.2.2 shall be subjected to the examinations and tests listed in Table II.

Table II

Examinations and Tests

Examination	Test	Requirement
Visual	4.3.1	3.3 and Section 5

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Table II (Continued)

Hardness, Shore D	4.3.2	3.2
Non-Extractables (of powder), percent	4.3.4	3.2
Screen Analysis (of powder), percent passing through a U.S. Number 200 sieve	4.3.5	3.2

4.3 Inspection methods.

4.3.1 Visual examination. Visually examine all containers for conformance to 3.3 and Section 5.

4.3.2 Hardness. Hardness shall be determined in accordance with ASTM D 2240 with the following exception:

a. Thoroughly stir the liquid before removal from its container. Weigh a clean, polytetrafluoroethylene container. Weigh into the container 100 plus or minus 0.5 parts by weight of powder and 28.5 plus or minus 0.5 parts by weight of liquid. Mix the ingredients until a uniform consistency is obtained. Let the mixture set up at 21 to 32 degrees Celsius (C) (70 to 90 degrees Fahrenheit (F)) for 120 plus or minus 5 minutes prior to use. Cure the specimens at a temperature of 97 to 103°C (207 to 217°F) for 120 to 125 minutes. Increase the temperature to 120 to 130°C (248 to 266°F) for a minimum of 16 hours. Manually remove the specimens from the mold. Condition prior to test for a minimum of 2 hours at 21 to 32°C (70 to 90°F).

4.3.3 Furfural. Furfural shall be determined on the liquid in accordance with the following:

4.3.3.1 Reagent. Saturated solution of 2, 4-dinitrophenylhydrazine

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Weigh 4.0 grams, to the nearest milligram (mg), of 2, 4-dinitrophenylhydrazine and add to a 1000 milliliter (ml) volumetric flask. Mix in 2 Normal (N) hydrochioric acid (HCl) that has been cooled to 0°C (32°F). Dilute to 1000 ml with cooled 2N HCl and allow to stand for a minimum of 16 hours at ambient temperature.

4.3.3.2 Procedure. Weigh 0.04 grams of the sample, to the nearest 0.1 mg, in a 250 Erlenmeyer flask having a ground glass stopper. Add 100 ml of reagent and stopper immediately. Allow to stand in an ice water bath for 60 to 65 minutes with periodic shaking. Filter the precipitate through a tared medium porosity sintered glass crucible. Wash with 2N HCl and then wash with distilled water. Dry in an oven at 95 to 105°C (203 to 221°F) for a minimum of 2 hours. Calculate the percent furfural as follows:

$$\text{Percent furfural} = \frac{(A - B)(0.348)}{C} \times 100$$

where:

A = Weight of crucible and contents

B = Weight of crucible

C = Weight of sample

4.3.4 Non-Extractables. Weigh approximately 1.0 gram, to the nearest mg of the carbonaceous powder into a tared glass extraction thimble. Add 50 ml of 95 percent ethyl alcohol to a 100 ml Soxhlet extraction apparatus. Set up the apparatus to reflux for a minimum of 16 hours. The extraction apparatus should cycle every 3 to 5 minutes. At the end of the reflux period remove and dry the thimble and residue in a 100°C (212°F) oven for a minimum of 2 hours Remove from the oven, allow to cool in a desiccator and weigh. Calculate the percent non-extractables as follows:

$$\text{Percent non-extractables} = \frac{B - A}{C} \times 100$$

where:

A = Tare weight of thimble

B = Weight of thimble and residue after extraction

C = Weight of sample

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4.3.5 Screen analysis. Weigh a 10 plus or minus 0.1 gram sample to the nearest mg. Add the material to a U.S. Number 200 sieve conforming to RR-S-366. Sift the material utilizing a nylon brush to disperse the particles. Empty the contents from the receiving pan onto tared glazed paper. Calculate the percent passing through the U.S. Number 200 sieve as follows:

$$\text{Percent through a 200 sieve} = \frac{A}{B} \times 100$$

where:

A = Weight from receiver pan

B = Original weight

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Level C. Unless otherwise specified by the procuring activity, the packaging and packing shall be in accordance with the best accepted commercial practice.

5.2 Marking. Marking shall be in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Title, number, and revision letter of this specification.

b. Material trade name.

c. Manufacturer's name.

d. Lot number, batch number, and date of manufacture.

e. Shelf life or storage limitations.

f. Net weight or volume.

6. NOTES

6.1 Intended use. The carbonaceous cement herein is intended for use in rocket motors.

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6.2 Ordering data. Procurement documents should include, but not be limited to, the following information:

a. Title, number, and revision letter of this specification.

b. Lot size, if applicable.

c. Place of delivery.

d. Size of containers.

e. Place of inspection.

f. Disposition of test data.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS 13019, dated 18 February 1965.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 5640-A028

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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Commanding General
U. S. Army Missile Command
ATTN: AMSMI-RCS
Redstone Arsenal, Alabama 35809

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